QUADRILATERAL

चतुर्भुज

PRO LEVEL SHEET

QUADRILATERAL (चतुर्भुज)

Pro Level Sheet

1. In a circle, ABCD is a cyclic quadrilateral in which AE is drawn parallel to CD, and BA is produced to F. If ∠ABC = 85° and ∠FAE = 24°, find the value of ∠BCD.

एक वृत्त में, ABCD एक चक्रीय चतुर्भुज है जिसमें AE को CD के समानांतर खींचा गया है, और BA को F तक बढ़ाया गया है। यदि ∠ABC = 85° और ∠FAE = 24° हो, तो ∠BCD का मान ज्ञात करें।

(a) 125° (b) 115°
(c) 124° (d) 119°

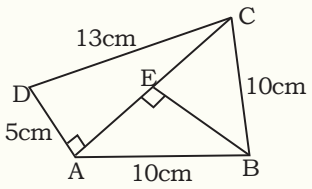
2. In a quadrilateral ABCD, the bisectors of ∠C and ∠D meet at point E. If ∠CED = 57° and ∠A = 47°,then the measure of ∠B is:

एक चतुर्भुज ABCD में, ∠C और ∠D के समद्विभाजक बिंदु E पर मिलते हैं। अगर ∠CED = 57° और ∠A = 47° है, तो ∠B का माप बताइए।

(a) 47° (b) 67°
(c) 77° (d) 57°

3. What is the area of a quadrilateral ABCD, (shown below) in which sides AB and BC are equal, sides AD and CD are of lengths 5 cm and 13 cm, respectively, and side AD is perpendicular to the diagonal AC?

उस चतुर्भुज ABCD का क्षेत्रफल क्या है, (नीचे दर्शाया गया है) जिसमें भुजाएं AB और BC समान हैं, भुजाएं AD और CD की लंबाई क्रमशः 5 सेमी और 13 सेमी हैं, और भुजा AD, विकर्ण AC के लंबवत है?



(a) 75cm² (b) 78cm²
(c) 82cm² (d) 80cm²

4. In a quadrilateral ABCD, AB = BC, AD = DC. ∠ABD = 68°, ∠ADB = (2y – 7)°, ∠BDC = 33°, ∠DBC = (3x + 2)°. Then the value of 2x + 3y is:

एक चतुर्भुज ABCD में AB = BC, AD = DC. ∠ABD = 68°, ∠ADB = (2y – 7)°, ∠BDC = 33°, ∠DBC = (3x + 2)° हैं। तो 2x + 3y का मान क्या है?

(a) 108° (b) 144°
(c) 104° (d) 118°

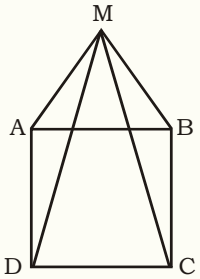
5. In a square ABCD,E is a point inside the square such that ΔDEC is an equilateral triangle. If E is joined to vertices A and B of the square, what is the degree measure of ∠AEB?

एक वर्ग ABCD में, E वर्ग के अंदर एक बिंदु है जैसे कि ΔDEC एक समबाहु त्रिभुज है। यदि E को वर्ग के शीर्ष A और B से जोड़ दिया जाए, तो ∠AEB का डिग्री माप क्या है?

(a) 135° (b) 150°
(c) 210° (d) 225°

6. ABCD is a square and ΔMAB is an equilateral triangle. MC and MD are joined. What is the degree measure of ∠MDC?

ABCD एक वर्ग है और ΔMAB एक समबाहु त्रिभुज है। MC और MD जुड़े हुए हैं। ∠MDC का अंश माप कितना होगा?



(a) 78° (b) 60°
(c) 65° (d) 75°

7. PQ is parallel to SR in a trapezium PQRS. It is given that PQ > SR and the diagonals PR and QS intersect at O. If PO = 3x – 15, OQ = x + 9, OR = x – 5 and OS = 5 and x has two values x_1 and x_2 then the value of $(x_1^2 - x_2^2)$ is:

एक समलंब चतुर्भुज PQRS में PQ, SR के समानांतर है। यह दिया गया है कि PQ > SR है तथा विकर्ण PR और QS, O पर प्रतिच्छेद करते हैं। यदि PO = 3x – 15, OQ = x + 9, OR = x – 5 और OS = 5 है तथा x के दो मान x_1 और x_2 हैं, तो $(x_1^2 - x_2^2)$ का मान क्या है?

(a) 15 (b) 11
(c) 19 (d) 13

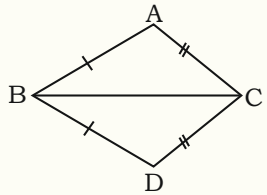
8. In a square ABCD, diagonals AC and BD intersect at O. The angle bisector of ∠CAB meets BD and BC at F and G, respectively. OF : CG is equal to:

एक वर्ग ABCD में, विकर्ण AC और BD एक-दूसरे को बिंदु O पर काटते हैं। कोण ∠CAB का कोण समद्विभाजक क्रमशः BD और BC को बिंदु F और G पर मिलता है। OF : CG बराबर है:

(a) 1 : 2 (b) 1 : 3
(c) $1 : \sqrt{2}$ (d) $1 : \sqrt{3}$

9. In the given figure, AB = DB & AC = DC. If angle ∠ABD = 58° and ∠DBC = (2x - 4)°, ∠ACB = (y + 15)° and ∠DCB = 63°, then the value of 2x + 5y is:

दी गई आकृति में, AB = DB और AC = DC है। यदि कोण ∠ABD = 58° और ∠DBC = (2x - 4)°, ∠ACB = (y + 15)° और ∠DCB = 63° है, तो 2x + 5y का मान क्या है?



(a) 325° (b) 273°
(c) 259° (d) 268°

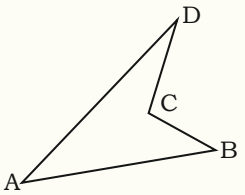
10. WXYZ is a quadrilateral in which XQ and ZP are perpendicular to WY such that XQ = ZP. Diagonals ZX and WY intersect each other at point O. If OX = 12 cm, then find the value of ZX (in cm).

WXYZ एक चतुर्भुज है जिसमें XQ और ZP, WY पर इस प्रकार लंबवत हैं कि XQ = ZP है। विकर्ण ZX और WY एक दूसरे को बिंदु O पर प्रतिच्छेद करते हैं। यदि OX = 12 सेमी है, तो ZX का मान (सेमी में) ज्ञात कीजिए।

(a) 12 (b) 24
(c) 36 (d) 18

11. If in the following figure (not to scale), ∠DAB + ∠CBA = 90°, BC = AD, AB = 20cm, CD = 10cm then the area of the quadrilateral ABCD is:

यदि निम्नलिखित आकृति में (पैमाने पर नहीं), ∠DAB + ∠CBA = 90°, BC = AD, AB = 20 सेमी, CD = 10 सेमी तो चतुर्भुज ABCD का क्षेत्रफल है:



(a) 120 cm^2 (b) 150 cm^2
(c) 100 cm^2 (d) 75 cm^2

12. In a quadrilateral ABCD, E is a point in the interior of the quadrilateral such that DE and CE are the bisectors of ∠D and ∠C, respectively. If ∠B = 82° and ∠DEC = 80°, then ∠A = ?

एक चतुर्भुज ABCD में, E चतुर्भुज के अभ्यंतर में एक बिंदु है जिससे DE और CE क्रमशः ∠D और ∠C के समद्विभाजक हैं। यदि ∠B = 82° और ∠DEC = 80° है, तो ∠A = ?

(a) 84° (b) 78°
(c) 75° (d) 81°

13. The side MN of a parallelogram MNOP is produced to Q such that MN = NQ. PQ intersects ON at R. The point R divides ON in the ratio:

एक समांतर चतुर्भुज MNOP की भुजा MN को Q तक इस प्रकार बढ़ाया गया है कि MN = NQ है। PQ, ON को R पर प्रतिच्छेद करती है। बिंदु R, ON को इस अनुपात में विभाजित करता है:

(a) 3 : 1 (b) 2 : 1
(c) 1 : 2 (d) 1 : 1

14. M is the mid-point of side QR of a parallelogram PQRS (P being on the top left hand, followed by other points going clockwise). The line SM is drawn intersecting PQ produced at T. What is the length (in terms of the length of SR) of PT?

M एक समांतर चतुर्भुज PQRS की भुजा QR का मध्य-बिंदु है (P ऊपरी बाएँ हाथ पर है, जिसके बाद अन्य बिंदु दक्षिणावर्त दिशा में हैं)। रेखा SM बढ़ाई गई PQ को T पर प्रतिच्छेद करती है। PT की लंबाई (SR की लंबाई के संदर्भ में) क्या है?

(a) 2SR (b) SR
(c) $\frac{3SR}{2}$ (d) $\frac{SR}{2}$

15. The lengths of two sides of a parallelogram are 3 cm and 10 cm. What is the sum of the squares of the diagonals of the parallelogram?

एक समांतर चतुर्भुज की दो भुजाओं की लंबाई 3 सेमी और 10 सेमी है। समांतर चतुर्भुज की विकर्णों के वर्गों का योग क्या है?

(a) 218 cm^2 (b) 169 cm^2
(c) 206 cm^2 (d) 109 cm^2

16. A square and a rhombus have the same base and the rhombus is inclined at 30". What is the ratio of area of the square to the area of the rhombus:

एक वर्ग और एक समचतुर्भुज का आधार समान है और समचतुर्भुज 30" पर झुका हुआ है। वर्ग के क्षेत्रफल का समचतुर्भुज के क्षेत्रफल से अनुपात क्या है।

(a) $\sqrt{2}:1$ (b) 2 : 1
(c) 1 : 1 (d) $2:\sqrt{3}$

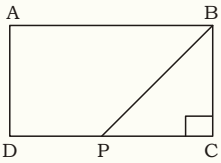
17. ABCD is a square, A is joined to a point P on BC and D is joined to a point Q on AB. If AP = DQ and AP intersects DQ at R then ∠DRP is:

ABCD एक वर्ग है, A को BC पर स्थित एक बिंदु P से जोड़ा गया है और D को AB पर स्थित एक बिंदु Q से जोड़ा गया है। यदि AP = DQ और AP, DQ को R पर प्रतिच्छेद करता है तो ∠DRP कितना होगा?

(a) 60° (b) 120°
(c) 90° (d) Can't be determined

18. Find the area of ABPD if AB = DC = 4 m, AD = BC = 2.5 m, and PC = 1 m.

ABPD का क्षेत्रफल ज्ञात करें यदि AB = DC = 4 मी, AD = BC = 2.5 मी और PC = 1 मी है।



(a) 9.0 m^2 (b) 7.85 m^2
(c) 8.75 m^2 (d) 6.25 m^2

19. ABCD is a square. X is the mid-point of AB. Y is a point on BC such that BY = $\frac{1}{3}$ BC. If the area of ΔBXY is equal to 300 cm^2, then find the length of diagonal AC.

ABCD एक वर्ग है। X, AB का मध्य-बिंदु है। Y, BC पर एक ऐसा बिंदु है कि BY = $\frac{1}{3}$ BC है। यदि ΔBXY का क्षेत्रफल 300 सेमी² के बराबर है, तो विकर्ण AC की लंबाई ज्ञात कीजिए।

(a) $50\sqrt{2}$ (b) $60\sqrt{2}$
(c) 60 (d) 50

20. The area of a rhombus is 24 m^2 and the length of one of its diagonals is 8 m. What will be the length of each side of the rhombus?

एक समचतुर्भुज का क्षेत्रफल 24 मी² और इसके एक विकर्ण की लंबाई 8 मी है। समचतुर्भुज की प्रत्येक भुजा की लंबाई कितनी होगी?

(a) 6 m (b) 4 m
(c) 10 m (d) 5 m

21. The area of the square formed on the diagonal of a rectangle as its side is $108\frac{1}{3}$ % more than the area of the rectangle. If the perimeter of the rectangle is 28 units, find the difference between the sides of the rectangle?

एक आयत के विकर्ण पर बने वर्ग का क्षेत्रफल उस आयत के क्षेत्रफल से $108\frac{1}{3}\%$ अधिक है। यदि आयत का परिमाप 28 इकाई है, तो आयत के दोनों भुजाओं के बीच अंतर ज्ञात करें?

(a) 4 (b) 6
(c) 2 (d) 8

22. The measures of adjacent sides of a parallelogram are 4a and 3a. If the angle between them is 60°, then the value of one of the diagonals of the parallelogram will be equal to which of the following?

एक समांतर चतुर्भुज की आसन्न भुजाओं के माप 4a और 3a हैं। यदि उनके बीच का कोण 60° है, तो समांतर चतुर्भुज के किसी एक विकर्ण का मान इनमें से किसके बराबर होगा?

(a) $\sqrt{13}a$ (b) $2\sqrt{3}a$
(c) $5\sqrt{3}a$ (d) $3\sqrt{3}a$

23. P is the midpoint of side BC of parallelogram ABCD such that ∠BAP = ∠DAP is. If AD = 10 cm, then CD = ?

P समांतर चतुर्भुज ABCD की भुजा BC का मध्यबिंदु इस प्रकार है कि ∠BAP = ∠DAP है। यदि AD = 10 सेमी तो CD = ?

(a) 10 cm (b) 8 cm
(c) 5 cm (d) 6 cm

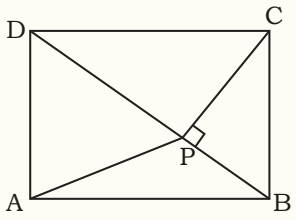
24. In a parallelogram ABCD, the length of the line joining the midpoints of AD and AC is 2 units. If the perimeter of the parallelogram is 26 units, then find the length of AD.

समांतर चतुर्भुज ABCD में AD और AC के मध्यबिंदुओं को मिलाने वाली रेखा की लंबाई 2 इकाई है। यदि समांतर चतुर्भुज का परिमाप 26 इकाई है, तो AD की लंबाई ज्ञात कीजिए

(a) 8 units (b) 4 units
(c) 6 units (d) 9 units

25. In the given figure ABCD is a rectangle, if DP = 8cm, PB = 2cm and PC⊥DB, then find AP?

दिए गए चित्र में, ABCD एक आयत है, यदि DP = 8cm, PB = 2cm और PC ⊥ DB है, तो AP की लंबाई होगी।



(a) $4\sqrt{6}$ cm (b) $2\sqrt{13}$ cm
(c) 6 cm (d) $\sqrt{57}$ cm

26. If the sum of the diagonals of a rhombus is L and the perimeter is 4P, find the area of the rhombus?

यदि एक समचतुर्भुज के विकर्णों का योग L और उसका परिमाप 4P है, तो उस समचतुर्भुज का क्षेत्रफल क्या होगा?

(a) $\frac{1}{4}(L^2 - P^2)$ (b) $\frac{1}{4}(L^2 - 4P^2)$
(c) $\frac{1}{2}(L^2 - 4P^2)$ (d) $\frac{1}{4}(L^2 - 3P^2)$

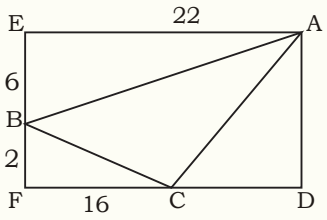
27. In a trapezium, the two non-parallel sides are equal in length, each being of 5 cm. The parallel sides are at a distance of 3 cm apart. If the smaller side of the parallel sides is of length 2 cm, then the sum of the diagonals of the trapezium is.

एक समलंब में, दो असमांतर भुजाएँ लंबाई में बराबर हैं, प्रत्येक की लंबाई 5 सेमी है। समानांतर भुजाएँ 3 सेमी की दूरी पर हैं। यदि समांतर भुजाओं की छोटी भुजा की लंबाई 2 सेमी है, तो समलम्ब के विकर्णों का योग है।

(a) $10\sqrt{5}$ cm (b) $6\sqrt{5}$ cm
(c) $3\sqrt{5}$ cm (d) $5\sqrt{5}$ cm

28. In the given figure, EADF is a rectangle and the vertices of triangle ABC lie on the sides of the rectangle. AE = 22, BE = 6, CF = 16 and BF = 2. Then the length of the line joining the midpoints of sides AB and BC will be-

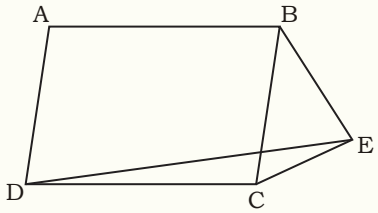
दिए गए चित्र में, EADF एक आयत है और त्रिभुज ABC के शीर्ष आयत की भुजाओ पर स्थित है। AE = 22, BE = 6, CF = 16 और BF = 2 है। तब भुजा AB और BC के मध्य बिन्दुओ को मिलाने वाली रेखा की लंबाई होगी–



(a) $4\sqrt{2}$ (b) 5
(c) 3.5 (d) 4.8

29. In the given figure, ABCD is a rhombus and BCE is an isosceles triangle, with BC = CE, ∠CBE = 84° and ∠ADC = 78°, then what is the value (in degrees) of ∠DEC?

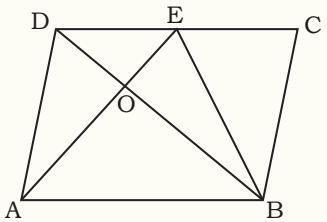
दी गई आकृति में, ABCD एक समचतुर्भुज है तथा BCE एक समद्विबाहु त्रिभुज है, जिसमें BC = CE, ∠CBE = 84° तथा ∠ADC = 78° है, तो ∠DEC का मान (डिग्री में) क्या है?



(a) 20 (b) 28
(c) 33 (d) 36

30. ABCD is a parallelogram as shown is figure ar (ΔDOE) = 7cm² and ar (ΔAOB) = 63cm², then find the area of parallelogram.

ABCD एक समान्तर चतुर्भुज है जिसमें क्षेत्रफल (ΔDOE) = 7 cm² और क्षेत्रफल (ΔAOB) = 63 cm² है, तो समान्तर चतुर्भुज का क्षेत्रफल ज्ञात कीजिये।



(a) 164 cm² (b) 147 cm²
(c) 84 cm² (d) 168 cm²

31. ABCD is a plate in the shape of a parallelogram. EF is the line parallel to DA and passing through the point of intersection O of the diagonals AC and BD. Further, E lies on DC, and F lies on AB. The triangular portion DOE is cut out from the plate ABCD. What is the ratio of area of the remaining portion of the plate to the whole?
ABCD एक समांतर चतुर्भुज के आकार की एक प्लेट है। EF, DA के समानांतर रेखा है और विकर्ण AC और BD के प्रतिच्छेदन बिंदु O से होकर गुजरती है। इसके अलावा E, DC पर स्थित है, और F, AB पर स्थित है। त्रिभुजाकार भाग DOE को प्लेट ABCD से काट दिया जाता है। प्लेट के शेष भाग के क्षेत्रफल का संपूर्ण क्षेत्रफल से अनुपात क्या है?

(a) $\frac{5}{8}$ (b) $\frac{5}{7}$

(c) $\frac{3}{4}$ (d) $\frac{7}{8}$

32. Side AB of rectangle ABCD is divided into four equal parts by points X, Y, Z the ratio of the.

$\frac{\text{Area of } \Delta XYD}{\text{Area of Rectangle ABCD}}$ is

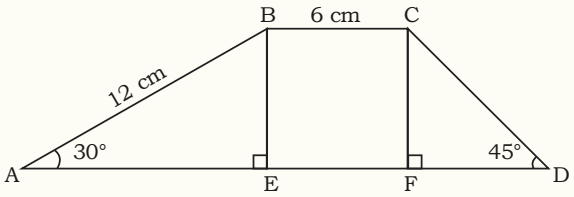
आयत ABCD की भुजा AB को बिंदु X, Y, Z के अनुपात से चार बराबर भागों में विभाजित किया गया है।

$\frac{\text{Area of } \Delta XYD}{\text{Area of Rectangle ABCD}}$

(a) $\frac{1}{7}$ (b) $\frac{1}{6}$

(c) $\frac{1}{9}$ (d) $\frac{1}{8}$

33. In a trapezium ABCD, $\angle BAE = 30°$, $\angle CDF = 45°$, BC = 6 cm and AB = 12 cm. Find the area of ABCD.
एक समलंब ABCD में, $\angle BAE = 30°$, $\angle CDF = 45°$, BC = 6 सेमी और AB = 12 सेमी ABCD का क्षेत्रफल ज्ञात कीजिए।



(a) $18\left(3+\sqrt{3}\right)\text{cm}^2$ (b) $36\sqrt{3}\ \text{cm}^2$

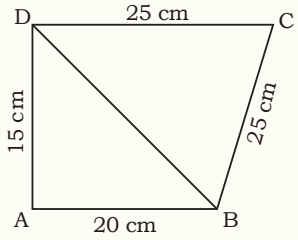
(c) $12\left(3+2\sqrt{3}\right)\text{cm}^2$ (d) None of these

34. Find the area of a trapezium ABCD in which AB || DC and AB = 26 cm, BC = 25 cm, CD = 40 cm and DA = 25 cm.
एक समलंब ABCD का क्षेत्रफल ज्ञात कीजिए जिसमें AB || DC और AB = 26 सेमी, BC = 25 सेमी, CD = 40 सेमी और DA = 25 सेमी।

(a) 648cm^2 (b) 792cm^2

(c) 660cm^2 (d) 978cm^2

35. In the given figure, AD = 15 cm, AB = 20 cm and BC = CD = 25 cm then what will be the area of ABCD.
दिये गये चित्र में, AD = 15 सेमी, AB = 20 सेमी तथा BC = CD = 25 सेमी हो तो ABCD का क्षेत्रफल क्या होगा।



(a) $\frac{25}{4}\left(24+25\sqrt{3}\right)\text{cm}^2$ (b) $24\left(25+24\sqrt{3}\right)\text{cm}^2$

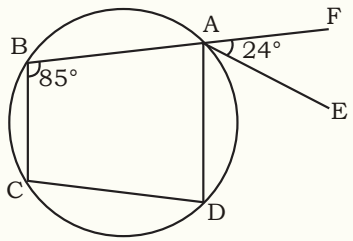
(c) $\frac{25}{2}\left(24+25\sqrt{3}\right)$ (d) None of these

ANSWER KEY

1.	(d)	2.	(b)	3.	(b)	4.	(c)	5.	(b)	6.	(d)	7.	(b)	8.	(c)	9.	(b)	10.	(b)
11.	(d)	12.	(b)	13.	(d)	14.	(a)	15.	(a)	16.	(b)	17.	(c)	18.	(c)	19.	(b)	20.	(d)
21.	(c)	22.	(a)	23.	(c)	24.	(d)	25.	(b)	26.	(b)	27.	(b)	28.	(b)	29.	(c)	30.	(d)
31.	(d)	32.	(d)	33.	(a)	34.	(b)	35.	(a)										

SOLUTIONS

1. (d)



We know,

$\angle ABC + \angle ADC = 180°$

$\angle ADC = 180° - 85° = 95°$

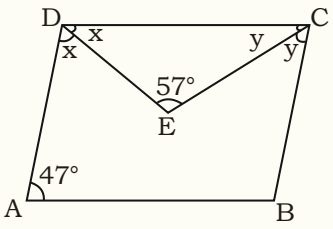
$\angle EAD = \angle ADC = 95°$ (Alternate interior angle of parallel lines CD and AE)

$\angle FAD = \angle FAE + \angle EAD$

$\angle FAD = 24° + 95° = 119°$

$\angle BCD = \angle FAD = 119°$ (Exterior angle of a cyclic quadrilateral)

2. (b)



Given,

DE and CE are angle bisectors of D and C respectively.

$\angle CED = 57°$

$\angle DAB = 47°$

Let $\angle EDC = \angle EDA = x°$ and $\angle ECD = \angle ECB = y°$

In ΔDEC,

$x + 57° + y = 180°$

$\Rightarrow x + y = 180° - 57° = 123°$...(i)

In quadrilateral ABCD

$\angle A + \angle B + \angle C + \angle D = 360°$

$\Rightarrow 47° + \angle B + 2y + 2x = 360°$

$\Rightarrow \angle B = 360° - 47° - 2(x + y)$

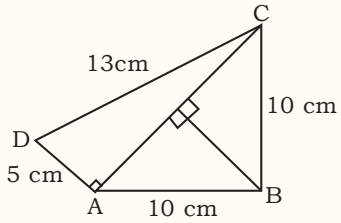
$= 360° - 47° - (2 \times 123°) = 67°$

SMART APPROACH:-

$\angle DEC = \frac{\angle A + \angle B}{2}$

$57° = \frac{47° + \angle B}{2} \Rightarrow B = 67°$

3. (b)



AC = 12 cm (triplet of 5, 12, 13)

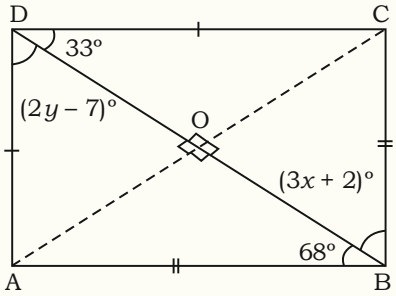
Area of Quadrilateral ABCD

= Area of ΔADC + Area of ΔABC

$= \frac{1}{2} \times 5 \times 12 + \sqrt{16 \times 6 \times 6 \times 4}$

$= 30 + 8 \times 6 = 30 + 48 = 78\ cm^2$

4. (c)



When adjacent side of quadrilateral are same then, diagonal of quadrilateral intersect each other at 90°

So, In ΔAOB, $\angle OAB = 180 - (68° + 90°) = 22°$

Then, In ΔACB, $\angle ACB = 22°$ [$\because$ AB = BC]

So, In ΔABC

$\angle A + \angle B + \angle C = 180°$

$\Rightarrow 22° + (68 + 3x + 2) + 22° = 180°$

$\Rightarrow x = 22°$

Now, In ΔDOC,

$\angle DOC + \angle ODC + \angle OCD = 180°$

$\Rightarrow 90° + 33° + \angle OCD = 180°$

$\Rightarrow \angle OCD = 57°$

So, $\angle OAD = 57°$ [$\because$ AD = DC]

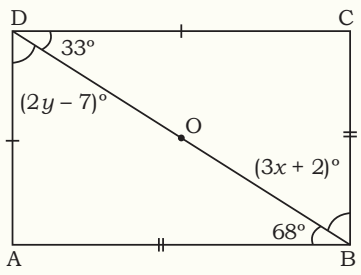
In ΔADC,

$57° + 57° + 33° + 2y - 7° = 180°$

$\Rightarrow y = 20$

$\therefore 2x + 3y = 2 \times 22 + 3 \times 20 = 104$

Alternate Method:



$\Delta ABD \cong \Delta CBD$

AB = BC

AD = DC

BD = BD ($\because$ SSS rule)

$\Rightarrow 2y - 7 = 33$

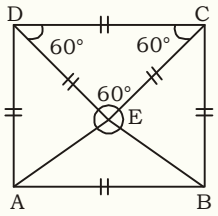
$y = 20°$

$\Rightarrow 3x + 2 = 68$

$x = 22°$

$\therefore 2x + 3y = 2 \times 22 + 3 \times 20 = 104°$

5. (b)



$\because \Delta DEC$ is an equilateral triangle and side of ΔDEC is also equal to side of square ABCD

Then, DC = CB = BA = AD = DE = CE

$\angle ADE = 90° - 60° = 30°$

Now, In ΔADE, AD = DE

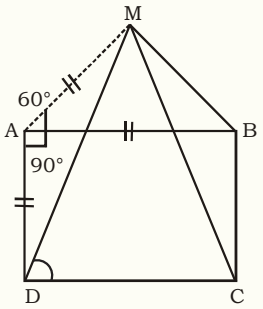
$\Rightarrow \angle DAE = \angle DEA = \frac{180° - 30°}{2} = 75°$

Similarly, $\angle BEC = 75°$

$\therefore \angle AEB = 360° - (2 \times 75° + 60°)$

$= 360° - 210° = 150°$

6. (d)



ABCD is a square.

$\Rightarrow$ AB = AD, AB = AM $\Rightarrow$ AD = AM

$\Rightarrow \angle ADM = \angle AMD$

In ΔADM,

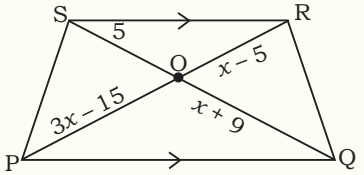
$\angle DAB + \angle BAM + \angle ADM + \angle AMD = 180°$

$\Rightarrow 90° + 60° + 2\angle ADM = 180°$

$\Rightarrow \angle ADM = 15°$ $\quad\quad$ $\because \angle ADC = 90°$

$\therefore \angle MDC = 90° - 15° = 75°$

7. (b)



If diagonal of a trapezium intersect each other then

$\frac{SO}{OQ} = \frac{RO}{OP} \Rightarrow \frac{5}{x+9} = \frac{x-5}{3x-15}$

$\Rightarrow 15x - 75 = x^2 - 5x + 9x - 45$

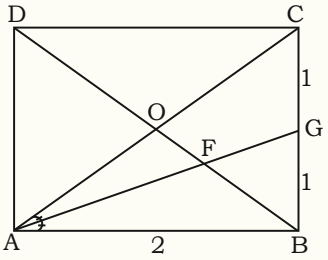
$\Rightarrow x^2 - 11x + 30 = 0$

$\Rightarrow x^2 - 6x - 5x + 30 = 0$

$\Rightarrow (x - 6)(x - 5) = 0 \Rightarrow x = 5, 6$

$\therefore x_1^2 - x_2^2 = 6^2 - 5^2 = 11$

8. (c)



Let, AB = 2

Since AB = BC hence F and G will be mid points of OB and BC respectively.

$\therefore$ CG = 1

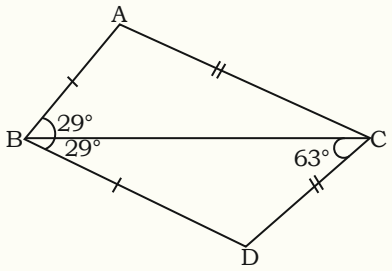
Now BD = $2\sqrt{2}$ = AC

OB = $\sqrt{2}$

OF = $\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$

OF : CG = $\frac{1}{\sqrt{2}} : 1 = 1 : \sqrt{2}$

9. (b)



As AB = DB, AC = DC and BC is common

$\therefore \Delta ABC \cong \Delta DBC$

$\therefore \Delta ABC \cong \Delta DBC = \frac{\angle ABD}{2} = \frac{58}{2} = 29°$

So, $\angle ABC = \angle DBC = 29°$

Now, $\angle ACB = \angle DCB = 63°$

Given, $\angle DBC = 29°$

$2x - 4 = 29°$

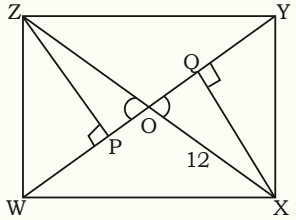
$2x = 33°$

$\therefore (y + 15) = 63°$

$\therefore y = 48°$

$2x + 5y = 33 + 5 \times 48 = 33 + 240 = 273°$

10. (b)



ZP = XQ (given)

$\angle ZOP = \angle XOQ$

(Vertically opposite angle)

and, $\angle ZPO = \angle XQO = 90°$

$\therefore \Delta ZPO \sim \Delta XQO$

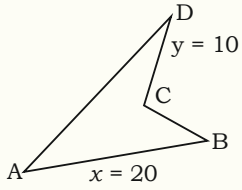
$\frac{ZP}{XQ} = \frac{OZ}{OX}$

$\frac{1}{1} = \frac{OZ}{12}$

$\therefore$ OZ = 12

ZX = 12 + 12 = 24 cm

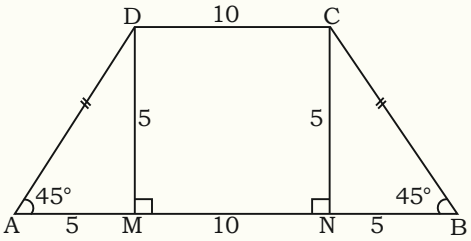
11. (d)



Area of quad. ABCD = $\frac{x^2 - y^2}{4}$ (Result)

= $\frac{20^2 - 10^2}{4} = \frac{400 - 100}{4}$ = 75 cm²

Alternate Method :-



$\angle DAB + \angle CBA = 90°$

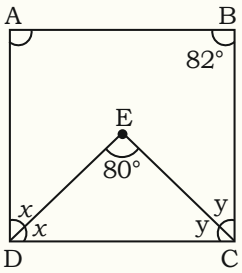
MN = 10

AM = BN = 5

DM = CN = 5

Area = $\frac{1}{2} \times (20 + 10) \times 5 = 75$ cm²

12. (b)



$\angle EDC = \frac{1}{2} \angle ADC$

$\angle ECD = \frac{1}{2} \angle BCD$

In ΔEDC

$\angle DEC + \angle EDC + \angle ECD = 180°$

$80° + x + y = 180°$

$x + y = 100°$

$\therefore 2(x + y) = 200°$

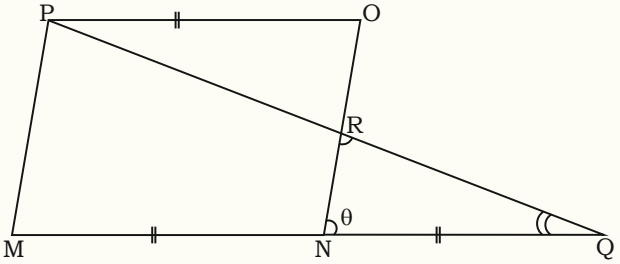
In $\square ABCD$

$\angle A + \angle B + \angle C + \angle D = 360°$

$\angle A + 82° + 2x + 2y = 360°$

$\angle A = 78°$

13. (d)



MQ | | PO

$\therefore \angle OPR = \angle NQR$ and

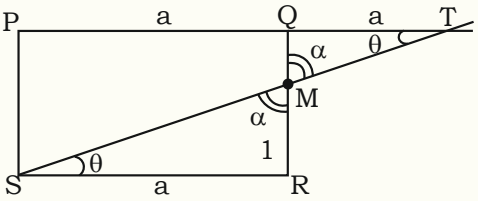
$\angle PON = \angle QNO$ {Transversal line properties}

$\therefore \Delta OPR \sim \Delta NQR$

$$\frac{OP}{NQ} = \frac{OR}{RN}$$

OR : RN = 1 : 1

14. (b)



PT is parallel to SR

$\therefore \angle PTS = \angle RST = \theta$

QM = MR ($\because$ M is the mid points)

$\angle QMT = \angle SMR$ (Vertically opposite angles)

$\therefore \Delta SMR \sim \Delta TMQ$

$$\frac{SR}{QT} = \frac{MR}{MQ} = \frac{SM}{MT} = \frac{1}{1}$$

$\therefore$ SR = QT

In parallelogram opposite sides are equal

$\therefore$ SR = PQ

Therefore, SR = PQ = QT

$\therefore$ PT = 2SR

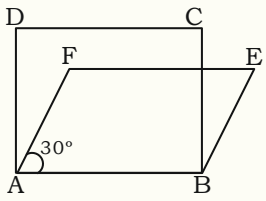
15. (a) In paralleogram

$(d_1)^2 + (d_2)^2$ = sum of the squares of all sides of parallelogram

$(d_1)^2 + (d_2)^2 = (10^2 + 3^2 + 10^2 + 3^2)$

$= (100 + 9 + 100 + 9) = 218\ cm^2$

16. (b)

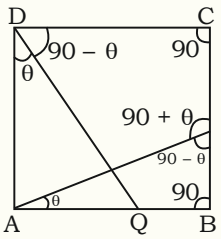


Area of square = a^2

Area of rhombus = $a \times a \times \sin 30° = \frac{a^2}{2}$

$$\frac{\text{Area of square}}{\text{area of rhombus}} = \frac{a^2}{\frac{a^2}{2}} = \frac{2}{1}$$

17. (c)



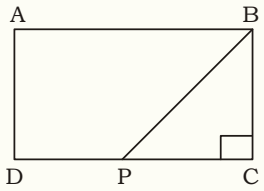
$\Delta ADQ \approx \Delta APB$

$\angle ADQ = \angle PAB = \theta$

$\angle DRP = 360 - (90 - \theta + 90 + 90 + \theta)$

$\angle DRP = 90$

18. (c)



AB = DC = 4m, PC = 1m,

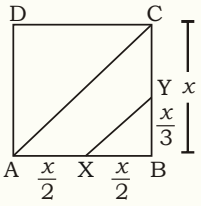
DP = (4 – 1) = 3m, BC = AD = 2.5m

Area of quadrilateral ABCD = $4 \times 2.5 = 10\ m^2$

Area of $\Delta PBC = \frac{1}{2} \times 2.5 \times 1 = 1.25\ m^2$

Area of quadrilateral ABDP = $(10 - 1.25) = 8.75\ m^2$

19. (b)



Side of square = x

Area of $\Delta BXY = \frac{1}{2} \times \frac{x}{2} \times \frac{x}{3} = \frac{1}{12}x^2$

Given,

$300 = \frac{1}{12}x^2 \Rightarrow x = 60$

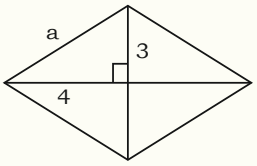
Then diagonal of square $= x\sqrt{2} = 60\sqrt{2}$ cm

20. (d) Area of rhombus $= \frac{1}{2} \times D_1 \times D_2$

$\Rightarrow 24 = \frac{1}{2} \times 8 \times D_2 \Rightarrow 24 = 4D_2$

$\Rightarrow D_2 = \frac{24}{4} = 6$

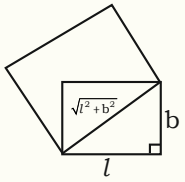
Since, the diagonals of a rhombus bisect each other at 90°.



Then the length of each side of the rhombus (a) is

$$= \sqrt{\left(\frac{8}{2}\right)^2 + \left(\frac{6}{2}\right)^2} = \sqrt{16+9} = \sqrt{25} = 5m$$

21. (c)



let the length of rectangle be l and bredth = b.

$108\frac{1}{3}\% = 1 + \frac{25}{300} = \frac{13}{12}$

ATQ,

$$\frac{\text{Area of square on diagonal}}{\text{Area of rectangle}} = \frac{25}{12}$$

$$\Rightarrow \frac{l^2 + b^2}{lb} = \frac{25}{12}$$

$(l + b)^2 = 25 + 24 = 49$

$\Rightarrow l + b = 7$ unit

$(l - b)^2 = 25 - 24 = 1$

$\Rightarrow l - b = 1$ unit

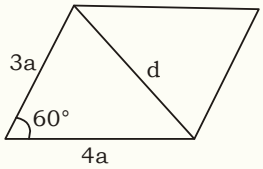
$2(l+b) = 28$ unit

7 unit $\rightarrow$ 14

1 unit $\rightarrow$ 2

Hence, $l - b = 2$

22. (a)

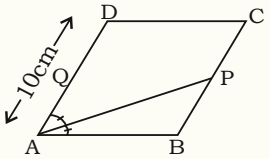


Using Cosine rule:- $d^2 = (3a)^2 + (4a)^2 - 2 \times 3a \times 4a \times \cos 60°$

$\Rightarrow d^2 = 9a^2 + 16a^2 - 12a^2$

$\Rightarrow d^2 = 13a^2 \Rightarrow d = \sqrt{13}a$

23. (c)



Given, $\angle BAP = \angle DAP$ and AD = 10 cm

ATQ, AD = BC and BP = PC = $\frac{AD}{2}$ = 5 cm

ABCD is a parallelogram

BP = PC (P is the midpoint of side BC)

$\Rightarrow \angle BAP = \angle DAP$ (given)

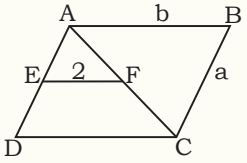
$\Rightarrow \angle DAP = \angle APB$ (alternate angle)

$\Rightarrow \angle BAP = \angle APB$

$\Rightarrow$ BA = BP (side opposite to the same angle)

$\therefore$ BA = BP = CD = 5cm (Opposite sides of a parallelogram are equal)

24. (d)



$\because$ EF || DC

By mid point theorem,

DC = 2 × EF

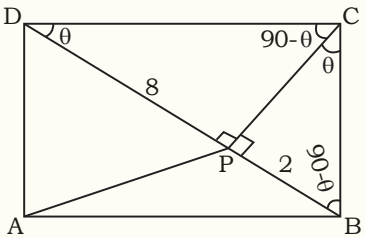
$\Rightarrow$ DC = 2 × 2 = 4 unit

Now, Perimeter of parallelogram = 2(a+b)

$\Rightarrow 26 = 2(a + 4) \Rightarrow a = 9$ unit

$\therefore$ The length of AD = 9 unit

25. (b)



In ΔDPC and ΔCPB

Similar Triangles

$\frac{CP}{8} = \frac{2}{CP}$

$CP^2 = 16$

CP = 4 cm

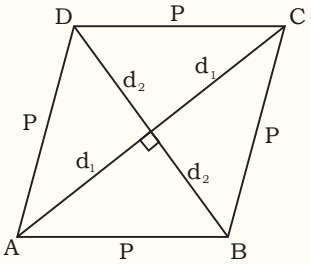
$(AP)^2 + (CP)^2 = (DP)^2 + (BP)^2$

$(AP)^2 + (4)^2 = (8)^2 + (2)^2$

$AP^2 = 52$

AP = $2\sqrt{13}$ cm

26. (b)



$\because$ $d_1^2 + d_2^2 = P^2$

$2d_1 + 2d_2 = L$

$d_1 + d_2 = \frac{L}{2}$

$(d_1 + d_2)^2 - 2d_1 d_2 = P^2$

$\left(\frac{L}{2}\right)^2 - 2d_1 d_2 = P^2$

$2d_1 d_2 = \frac{L^2}{4} - P^2$

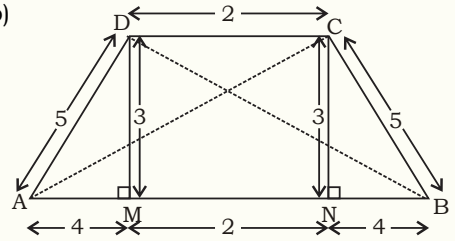
$2d_1 d_2 = \frac{L^2 - 4P^2}{4}$

$\because$ Area of Rhombus = $\frac{1}{2} \times 2d_1 \times 2d_2$

$= 2d_1 \times d_2$

Area $(2d_1 d_2) = \frac{1}{4}(L^2 - 4P^2)$

27. (b)



AM = $\sqrt{5^2 - 3^2}$ = 4 cm

In ΔANC

$AC^2 = AN^2 + NC^2$

$AC^2 = 36 + 9$

AC = $3\sqrt{5}$ cm

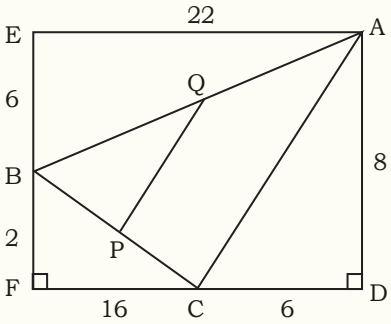
In ΔBMD –

$BD^2 = BM^2 + DM^2 = 36 + 9$

BD = $3\sqrt{5}$ cm

Sum of diagonals = AC + BD = $6\sqrt{5}$ cm

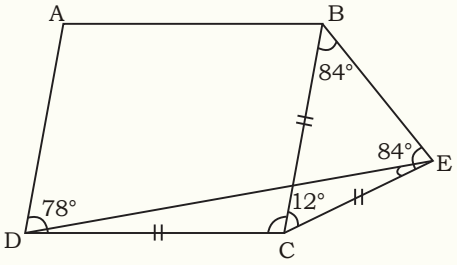
28. (b)



AE = FD = 22 cm
CD = 22 – 16 = 6 cm
EF = AD = 8 cm
In ΔACD, using pythagorean
Triplet (6, 8, 10)
AC = 10 cm

and $PQ = \frac{1}{2}AC = \frac{1}{2} \times 10 = 5\,cm.$ (Mid-point Theorem)

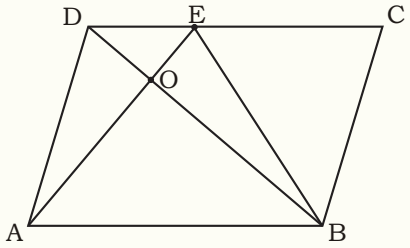
29. (c)



In ΔBCE:-
$\angle CBE = \angle BEC = 84°$
$\Rightarrow \angle BCE = 180° - 2 \times 84° = 12°$
Also, $\angle DCB = 180° - 78° = 102°$
$\therefore \angle DCE = 102° + 12° = 114°$
Also, DC = BC = CE
$\Rightarrow$ ΔDCE is also isoscales Δ
$\therefore \angle CDE = \angle CED$
Now,

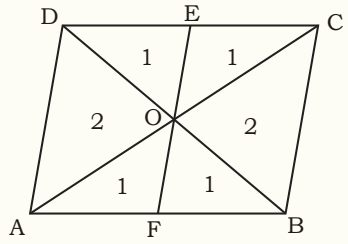
$$\angle DEC = \frac{180° - 114°}{2} = 33°$$

30. (d)



Consider, ΔABD, ΔBAE
They have same base and
same height and hence same area
$\therefore$ ar ΔABD – ar ΔAOB = ar ΔBAE – ar ΔAOB
$\Rightarrow$ ar ΔAOD = ar ΔBOE = x (say)
In trapezium ΔABED:-
ar ΔDOE × ar. ΔAOB = ar ΔDOA × ar. ΔEOB
$\Rightarrow 7 \times 63 = x^2$
$\Rightarrow x = 21$ = ar. AOD
$\therefore$ ar ΔADB = 21 + 63 = 84 cm²
$\therefore$ ar ABCD = 2 × 84 = 168 cm²

31. (d)

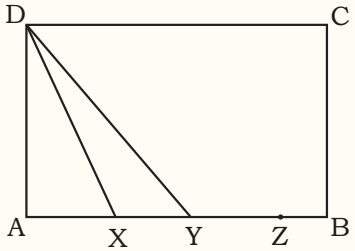


Let the complete area of parallelogram be 8 units diagonal BD will divide it in 2 equal parts
So, ar. ΔABD = ar. ΔBCD = 4 units
and CO divide ΔBCD in 2 equal parts
Now, EF||AD, So AD = EF
EO will divide ΔDOC in 2 equal parts

Area of triangle DOE = $\frac{2}{2}$ = 1 unit

Required ratio $= \frac{8-1}{8} = \frac{7}{8}$

32. (d)

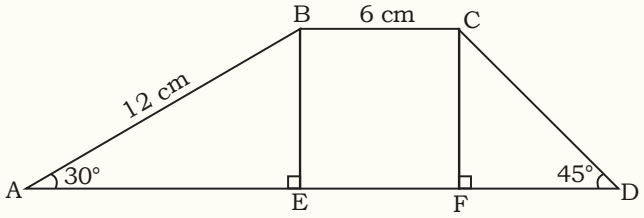


Ar. ABCD = AB × AD
Ar. ΔXYD:-

$$= \frac{1}{2} \times (XY) \times (AD) = \frac{1}{2} \times \frac{AB}{4} \times AD = \frac{1}{8} \times AB \times AD$$

$$\therefore \frac{ar.\Delta XYD}{ar.ABCD} = \frac{1}{8}$$

33. (a)



In Trapezium ABCD,
In ΔABE,

$$\cos 30° = \frac{AE}{AB}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{AE}{12} \Rightarrow AE = 6\sqrt{3}$$

In ΔCFD,

$$\tan 45° = \frac{CF}{FD}$$

$$\Rightarrow 1 = \frac{6}{FD} \Rightarrow FD = 6 \text{ cm}$$

$\because$ BC = EF = 6 cm

Now the area of trapezium $= \frac{1}{2}$ × (Sum of parallel sides) × height

$$= \frac{1}{2} \times (BC + AD) \times BE$$

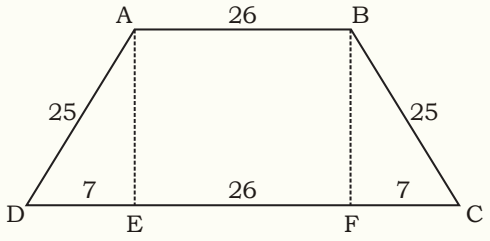
($\therefore$ AD = AE + EF + FD

$= 6\sqrt{3} + 6 + 6$)

$$= \frac{1}{2} \times (18 + 6\sqrt{3}) \times 6$$

Area $= 18(3 + \sqrt{3})\,cm^2$

34. (b)



Let height of parallelogram be h

Two perpendicular AE and BF are drawn on CD

$\therefore$ AB = EF = 26 cm

$\Rightarrow$ DE = FC = $\frac{1}{2}$ (40 – 26) = 7cm

In ΔAED,

$\therefore$ $(AD)^2 = (DE)^2 + (AE)^2$ (Pythagoras)

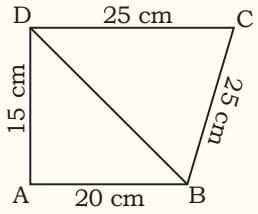
$\Rightarrow$ $(25)^2 = (7)^2 + h^2$

$\Rightarrow$ h = 24 cm

$\therefore$ Area of trapezium = $\frac{1}{2}$ × h × (Sum of parellel sides)

= $\frac{1}{2}$ × 24 × (26 + 40) = 792 cm^2

35. (a)



In right angled triangle DAB,

$BD^2 = AD^2 + AB^2$ (By Pythagoras theorem)

$(BD)^2 = (15)^2 + (20)^2$

BD = 25 cm

$\therefore$ Now BCD is an equilateral triangle.

Area = $\frac{\sqrt{3}}{4}(side)^2 = \frac{\sqrt{3}}{4}(25)^2$

Area of $\square$ABCD = ΔDAB + ΔBCD

= $\frac{1}{2}$ × Base × Height + $\frac{\sqrt{3}}{4}$ × 25 × 25

= $\frac{1}{2}$ × 15 × 20 + $\frac{\sqrt{3}}{4}$ × 25 × 25

= 150 + $\frac{\sqrt{3}}{4}$ × 25 × 25 = $\frac{25}{4}\left(24+25\sqrt{3}\right)$